

Practical Week 10

1. Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).

```
nocturnal@nocturnal:~/OS_Skill/Week-10$ echo "Hello from Sujit" > original.txt
nocturnal@nocturnal:~/OS_Skill/Week-10$ cat original.txt
Hello from Sujit
nocturnal@nocturnal:~/OS_Skill/Week-10$ ls -i original.txt
1724 original.txt
nocturnal@nocturnal:~/OS_Skill/Week-10$ stat original.txt
  File: original.txt
  size: 17          Blocks: 8          IO Block: 4096   regular file
Device: 8,48   Inode: 1724         Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/nocturnal)   Gid: ( 1000/nocturnal)
Access: 2026-09-27 13:28:36.785586740 +0000
Modify: 2026-09-27 13:28:17.725593980 +0000
Change: 2026-09-27 13:28:17.725593980 +0000
 Birth: 2026-09-27 13:27:52.601603114 +0000
nocturnal@nocturnal:~/OS_Skill/Week-10$ ln original.txt hardlink.txt
nocturnal@nocturnal:~/OS_Skill/Week-10$ ls -li original.txt hardlink.txt
1724 -rw-r--r-- 2 nocturnal nocturnal 17 Sep 27 13:28 hardlink.txt
1724 -rw-r--r-- 2 nocturnal nocturnal 17 Sep 27 13:28 original.txt
nocturnal@nocturnal:~/OS_Skill/Week-10$ echo "Modified using hard link" >> hardlink.txt
nocturnal@nocturnal:~/OS_Skill/Week-10$ cat original.txt
Hello from Sujit
Modified using hard link
nocturnal@nocturnal:~/OS_Skill/Week-10$ ln -s original.txt symlink.txt
nocturnal@nocturnal:~/OS_Skill/Week-10$ ls -li original.txt hardlink.txt symlink.txt
1724 -rw-r--r-- 2 nocturnal nocturnal 42 Sep 27 13:29 hardlink.txt
1724 -rw-r--r-- 2 nocturnal nocturnal 42 Sep 27 13:29 original.txt
1726 lrwxrwxrwx 1 nocturnal nocturnal 12 Sep 27 13:29 symlink.txt -> original.txt
nocturnal@nocturnal:~/OS_Skill/Week-10$ cat symlink.txt
Hello from Sujit
Modified using hard link
nocturnal@nocturnal:~/OS_Skill/Week-10$ ^C
nocturnal@nocturnal:~/OS_Skill/Week-10$ find . -inum 1724
./original.txt
./hardlink.txt
nocturnal@nocturnal:~/OS_Skill/Week-10$ rm original.txt
nocturnal@nocturnal:~/OS_Skill/Week-10$ cat hardlink.txt
Hello from Sujit
Modified using hard link
nocturnal@nocturnal:~/OS_Skill/Week-10$ cat symlink.txt
cat: symlink.txt: No such file or directory
nocturnal@nocturnal:~/OS_Skill/Week-10$
```

2. Develop a program that redirects standard input and output using file descriptors and dup2(). Demonstrate how shells implement I/O redirection.

```
Ubuntu × + ▾
GNU nano 8.7.1 src/io
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int input_fd, output_fd;

    input_fd = open("input.txt", O_RDONLY);

    if (input_fd < 0) {
        perror("input.txt");
        return 1;
    }

    output_fd = open("output.txt",
                    O_WRONLY | O_CREAT | O_TRUNC,
                    0644);

    if (output_fd < 0) {
        perror("output.txt");
        close(input_fd);
        return 1;
    }

    dup2(input_fd, STDIN_FILENO);
    dup2(output_fd, STDOUT_FILENO);

    close(input_fd);
    close(output_fd);

    char buffer[100];

    printf("This output is redirected to output.txt\n");
    printf("Reading from redirected input:\n");

    while (read(STDIN_FILENO, buffer, sizeof(buffer) - 1) > 0) {
        break;
    }

    return 0;
}
```

Output:

```
Ubuntu × + ▾
nocturnal@nocturnal:~/OS_Skill/Week-16$ nano src/io_redirection.c
nocturnal@nocturnal:~/OS_Skill/Week-16$ ^C
^[[200-gcc src/io_redirection.c -o bin/i^C
nocturnal@nocturnal:~/OS_Skill/Week-16$ gcc src/io_redirection.c -o bin/io_redirection
nocturnal@nocturnal:~/OS_Skill/Week-16$ ./bin/io_redirection
nocturnal@nocturnal:~/OS_Skill/Week-16$ cat output.txt
This output is redirected to output.txt
Reading from redirected input:
nocturnal@nocturnal:~/OS_Skill/Week-16$ cat input.txt
Hello Sujit! This is redirected input.
nocturnal@nocturnal:~/OS_Skill/Week-16$
```